

Phase 1: Research Ideation (1 Week)

 **Goal:** Understand existing work, finalize the tech stack.

 **Task Split:**

- **Person 1:** Research existing **lyric generation models** (GPT, LSTM, Markov).
- **Person 2:** Study **acoustic feature extraction** (Librosa, Wav2Vec2, Spotify API).
- **Person 3:** Explore **multi-modal learning techniques & dataset options**.

 **Output:** Finalized dataset, tech stack, and model selection.

Phase 2: Data Collection & Preprocessing (2 Weeks)

 **Goal:** Collect and preprocess lyrics + audio features.

 **Task Split:**

- **Person 1:** Scrape/download & clean **lyrics** (Musixmatch, Genius, Kaggle).
- **Person 2:** Extract **audio features** (tempo, pitch, key, spectral data).
- **Person 3:** **Align lyrics with audio embeddings**, normalize, and augment data.

 **Output:** Structured dataset (lyrics, audio features, metadata).

Phase 3: Model Design & Architecture (2 Weeks)

 **Goal:** Develop a multi-modal model for lyric generation.

 **Task Split:**

- **Person 1:** Implement **base text model** (GPT-2, T5, BART, LLaMA).
- **Person 2:** Experiment with **audio feature integration strategies**.
- **Person 3:** Implement **positional encoding adjustments** for rhythmic alignment.

 **Output:** Preliminary model + small-scale training results.

Phase 4: Training & Optimization (3 Weeks)

 **Goal:** Train a full-scale model & optimize performance.

 **Task Split:**

- **Person 1:** Implement **multi-modal training** (text + audio).

- **Person 2:** Optimize **loss function** (text coherence + rhythmic consistency).
- **Person 3:** Perform **hyperparameter tuning** (batch size, learning rate, dropout).

✅ **Output:** Fully trained model ready for evaluation.

Phase 5: Evaluation & Benchmarking (2 Weeks)

📌 **Goal:** Measure model performance & compare architectures.

👥 **Task Split:**

- **Person 1:** Evaluate using **text-based metrics** (BLEU, ROUGE, Perplexity).
- **Person 2:** Implement **music-specific evaluation** (syllable-beat alignment, genre consistency).
- **Person 3:** Conduct **human evaluation** (feedback from musicians/lyricists).

✅ **Output:** Best-performing model identified.

Phase 6: Deployment & Future Extensions (Final Week)

📌 **Goal:** Deploy the model with an interactive interface.

👥 **Task Split:**

- **Person 1:** Develop **API/web interface** (upload song, generate lyrics).
- **Person 2:** Implement **style/genre customization**.
- **Person 3:** Final testing, documentation, and presentation.

✅ **Final Output:** Deployed **AI Lyric Generator** 🚀 🎵